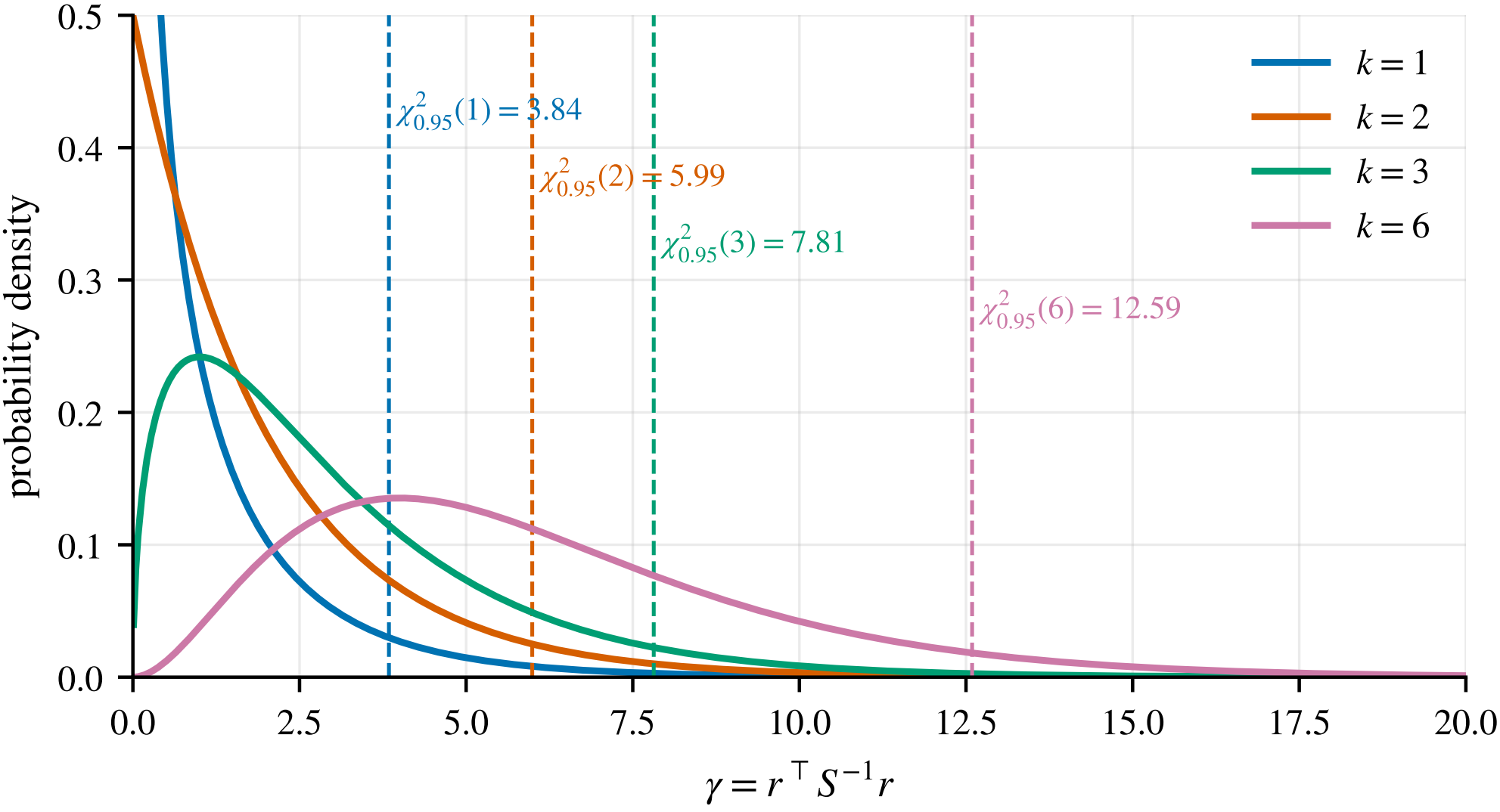
probability density
0.0
0.1
0.2
0.3
0.4
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
$\gamma = r^\top S^{-1} r$
$k = 1$
$k = 2$
$k = 3$
$k = 6$
$\chi^2_{0.95}(1) = 3.84$
$\chi^2_{0.95}(2) = 5.99$
$\chi^2_{0.95}(3) = 7.81$
$\chi^2_{0.95}(6) = 12.59$